

JONATHAN ZHU

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EDUCATION

University of Toronto

2024-2028

Honours B.Sc., Computer Science & Statistics, Software Engineering Stream **3.8 GPA**

- \$3000 renewable entrance scholarship

Relevant coursework: Intro to Linux programming, Data structures and algorithms

RELEVANT EXPERIENCE

Software Developer Intern — Royal Bank of Canada

May 2026 - Present

Python, Typescript, FastAPI, Langchain, Langgraph, Kubernetes, Apache Airflow

Toronto ON, Canada

- Selected out of **12,000+** applicants for the Amplify program to develop a **patent-pending** solution to address finance modernization challenges.
- Developed a explainable and transparent **LLM-driven** AutoML framework orchestrated using **Langgraph**, reducing **model iteration time** by **99%** and improving model **accuracy** by **88%** compared to existing options.
- Refactored Langchain tool calling mechanisms to use caching with time-to-live assessed based on average model iteration time, improving tool call **latency** by **73%**.

Embedded Systems Software Developer — aUToronto Student Club

Sept 2025 - May 2026

C++, CMake, Linux, ROS2

- Helped aUToronto earn **1st place overall** at the **2026 SAE AutoDrive Challenge™ II** by developing fault-tolerant ROS 2 infrastructure and embedded software for the team's autonomous vehicle.
- Implemented **ring buffer-based serial communication** in C++ to efficiently handle asynchronous data transfer between server and microcontroller, ensuring consistent data handling during high-frequency data transfer.
- Designed a lightweight **binary protocol** to communicate sensor health messages between server and microcontroller, enabling reliable low-latency status synchronization across embedded vehicle systems.

Backend Engineer Intern — PointClickCare

Sep 2025 - Dec 2025

Java, Kotlin, Springboot, SQL Server, Kubernetes, Redis, Terraform

Toronto ON, Canada

- Contributed to **3 core microservices** by optimizing APIs and implementing new features using **Spring Boot**, **Kubernetes** and **Terraform**, improving reliability and scalability.
- Achieved projected annual cloud **cost savings** of **\$3,000** from identifying and resolving a critical SQL Server database bottleneck by examining SQL Server Query Execution Plans, reducing **CPU utilization** by **36%**.
- Implemented a **Spring Boot** API retry system using Spring Retry library for a new patient responder service, improving fault tolerance and eliminated the occurrence of unresolved **transient errors** by **95%**.

Software Engineer Intern — Telesat

May 2025 - Aug 2025

C++, gRPC, Python, Selenium, Apache Kafka

Ottawa ON, Canada

- Designed and implemented an asynchronous **gRPC** file transfer **microservice** using **C++** to support fast and secure file transfers between management system and embedded devices.
- Automated GUI testing by writing **15+ scripts** using **Pytest** and **Selenium** that are integrated into Gitlab CI for automated component testing and continuous integration.
- Wrote custom **Selenium** tooling using **Pytest fixtures/hooks** and test tooling for **Apache Kafka** that will be used by **10+** developers estimated to save **~780** developer hours a year.

PROJECTS

Banter Mart - API Winer @ LA Hacks

React, Typescript, Supabase, Google Gemini

- Won the Best World API Integration prize among **1400+** participants at LA Hacks 2026 by integrating the World ID API to a agent-native online marketplace.
- Developed a full-stack commerce app using **React**, **TypeScript**, and **Supabase** that replaces traditional marketplace friction with an agentic negotiation interface, allowing buyers and sellers to align on prices effortlessly.

TECHNICAL SKILLS

Languages: Python, C, C++, Java, Rust, JavaScript, TypeScript, SQL, Protobuf

Frameworks/Libraries: React, Flask, Spring, SQLAlchemy, gRPC, Express.js, Selenium, Pytest, Django

Technologies: Git, Bash, Docker, AWS, Azure, Kubernetes, PostgreSQL, Apache Kafka, Terraform, Redis